higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING
NQF LEVEL 2

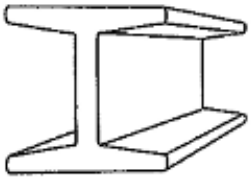
13 NOVEMBER 2019

This marking guideline consists of 7 pages.

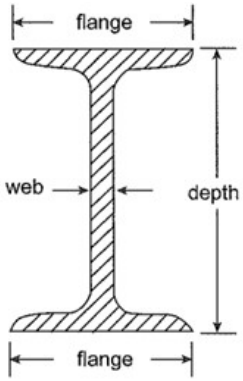
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QUESTION 1: INTRODUCTION TO CIVIL AND STRUCTURAL STEEL WORK

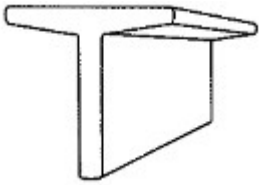
1.1 1.1.1



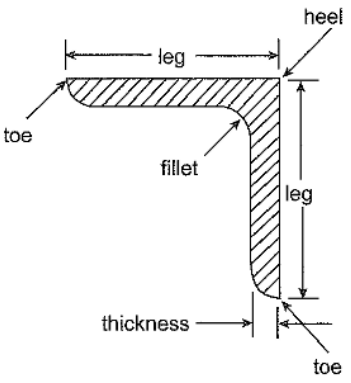
1.1.2



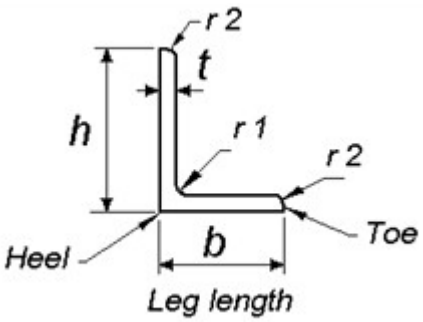
1.1.3



1.1.4



1.1.5

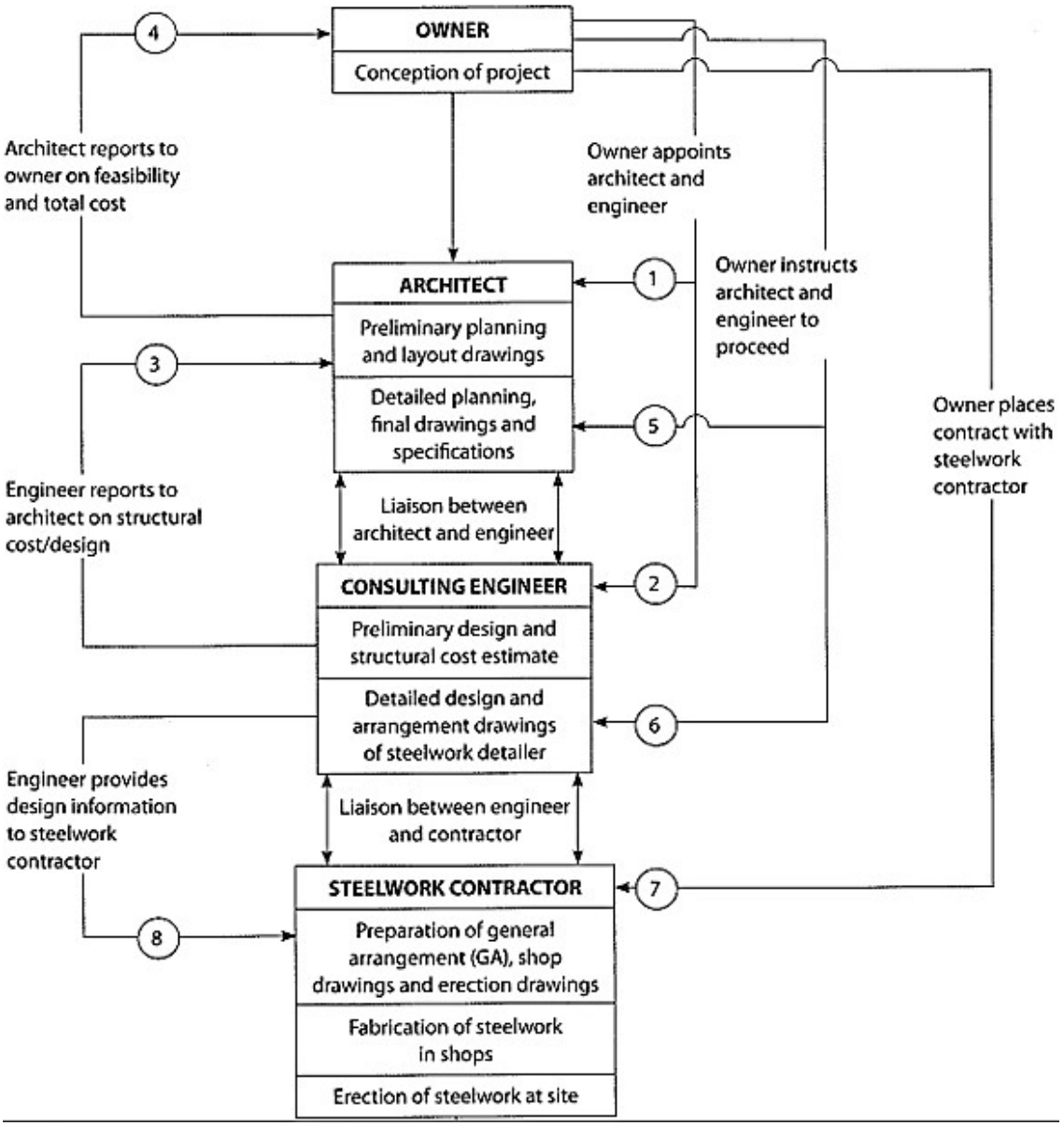


(5 × 2) (10)

1.2 1.2.1 True
1.2.2 False
1.2.3 True
1.2.4 True
1.2.5 True

(5 × 1) (5)

1.3



(2 for each role player and each role) (8)

1.4	1.4.1	B		
	1.4.2	L		
	1.4.3	A		
	1.4.4	C		
	1.4.5	H		
	1.4.6	D		
	1.4.7	K		
	1.4.8	J		
	1.4.9	G		
	1.4.10	F		
			(10 × 1)	(10)

1.5	1.5.1	Electric arc furnace		
	1.5.2	Concrete stairs/staircase/balustrade		
			(2 × 1)	(2)

1.6
- Quality of concrete
- Air entrainment
- Correct amount of steel
- Correct amount of cover
- Correct curing and consolidation
- Inhibitors and additives (Any 5 × 1) (5)

[40]

QUESTION 2: STRUCTURAL STEEL WORK AND REINFORCED CONCRETE

2.1 Reinforcing concrete improves its tensile resistance.✓ The high tensile strength of steel takes over as a resisting material at the bottom of the concrete.✓ (2)

2.2	2.2.1	High-yield deformed bar diameter 32 mm		
	2.2.2	Round mild-steel bar diameter 16 mm		
			(2 × 1)	(2)

2.3 Hot-rolled profiles: Steel members made from a billet of steel are heated up to red-hot and pushed through rollers, squeezing the metal into a particular shape.
Cold-rolled profiles: Structural members are formed by pushing steel through rollers without being preheated. (2 + 2) (4)

2.4 Workability is a property of freshly mixed concrete that determines its working characteristics,✓ i.e. the ease with which it can be mixed, placed, compacted and finished.✓ (2)

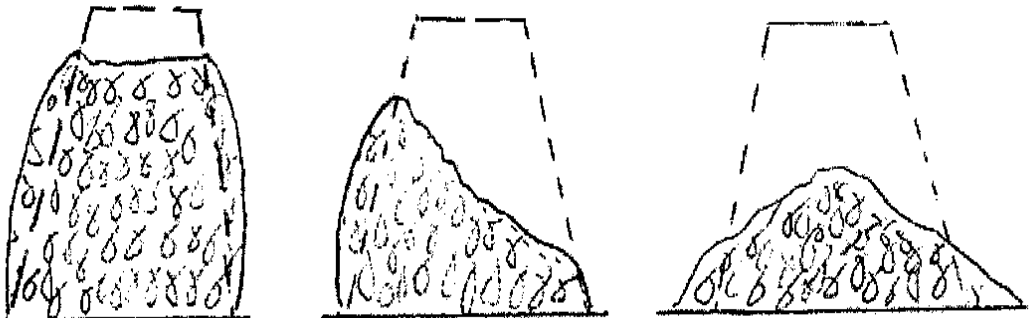
2.5
- Stadiums
- Bridges
- Skyscrapers
- Airports
- Gantries
- Factories (Any 5 × 1) (5)

2.6 Extensometer (1)

2.7 True (1)

2.8 Cube tests are used to determine the compressive strength✓ of a certain mix design.✓ (2)

2.9 t
r
u
e



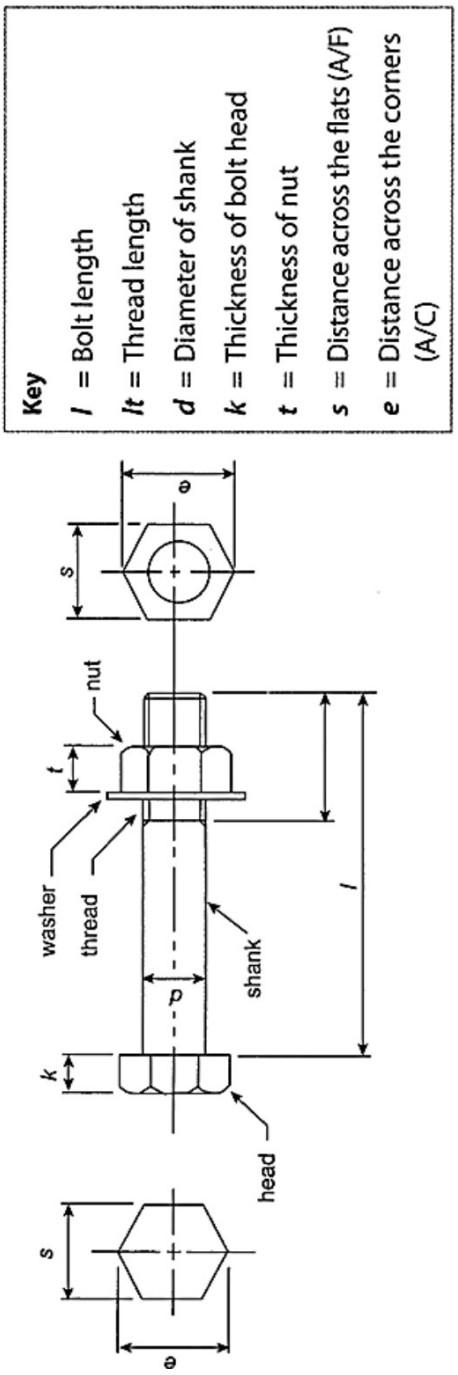
slump✓ shear slump✓ collapsed slump✓
(2 marks per correct sketch) (9)

2.10
- Slump test✓ to determine the workability of concrete✓
- Cube test✓ to determine the compressive strength of concrete✓ (2 × 2) (4)

2.11
- Strength
- Durability
- Fire resistance (3)

[35]

QUESTION 3: BOLTS IN STRUCTURAL STEEL WORK



MARKING CRITERIA	POSSIBLE MARK	ACHIEVED MARK
Accuracy (hex head, bolt, shank, threads)	4	
Line work	2	
Neatness	1	
Labels	3	

[10]

QUESTION 4: WELDING SYMBOLS AND JOINTS

4.1 False
4.2 True

(2 × 1) (2)

4.3
4.3.1 D
4.3.2 C
4.3.3 E
4.3.4 A
4.3.5 B

(5 × 1) (5)

4.4
- Flush contour
- Concave
- Convex

(3)

4.5
- Arrow
- Reference line
- Welding symbol
- Dimensions
- Tail

(5)
[15]

TOTAL: 100

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